Warm-up / Does liking an app prove learning?

Change your mind in proportion

Name: _____ Date: _____

Suggested lesson time: 30 minutes. Work individually, then compare when invited.

Useful words

Confidence: How strongly you believe a claim is true.

Relevant evidence: Information that bears on the particular claim being evaluated.

Random assignment: Using chance to decide which group receives a change, helping make groups comparable.

Read the scenario

Invented scenario: You are unsure whether a reading app improves comprehension. A friend says its colors are attractive. A report then describes a fair comparison in which app users answer more comprehension questions correctly than a similar group using usual materials; no details about the size of the difference are supplied.

Produce: Write how each report should affect confidence and one remaining question.

Your steps

1. Explain whether the comment about colors bears on comprehension.

2. Say whether the comparison should raise or lower confidence, relative to starting unsure.

3. Name one missing detail needed before making a strong claim.

Before you finish: Have you changed confidence for the learning result while keeping its limits visible?

Your response

Main activity / What should two studies change?

Name: _____

Read the scenario

Invented scenario: You begin unsure whether brighter streetlights reduce theft reports. Study A is one resident saying a street feels safer. Study B assigns 40 similar streets at random to brighter or unchanged lighting, keeps patrol policy the same, and records the same theft measure before and after: reports fall more in the brighter-light group. No exact effect size or uncertainty estimate is supplied.

Produce: Write one update for each study, a comparison of their strength, and one missing detail you would seek.

Your steps

1. For each study, say whether it should raise, lower, or leave your confidence roughly unchanged, and explain why.
2. Choose which should change confidence more, using the difference between personal impressions and a controlled comparison.
3. Name one missing detail you would seek before making a strong conclusion. Do not invent a precise percentage change in confidence.

Before you finish: Have you updated more for the stronger relevant evidence while keeping uncertainty about the size and reliability of the effect?

Your response

Exit ticket / Do reminders prevent missed deadlines?

Name: _____

Read the scenario

Invented scenario: You tentatively believe reminder messages reduce missed deadlines. Report A says one user likes the messages. Report B randomly assigns 200 students to reminders or no reminders for the same four-week course, uses the same deadline rules, and reports nearly equal missed-deadline rates. Exact rates and uncertainty estimates are not supplied.

Produce: Write how each report should change your confidence, which matters more, and one question still open.

Your steps

1. Judge how much the approval report bears on missed deadlines.

2. State the direction of the update from the comparison and explain why it matters more.

3. Name a missing detail and explain why the report does not prove reminders can never help.

Before you finish: Have you allowed contrary evidence to weaken the belief without turning one result into a universal verdict?

Your response